

RETAIL BUSINESS PERFORMANCE & PROFITABILITY ANALYSIS

1. Introduction

Retail businesses generate large amounts of transactional data related to customers, products, sales, discounts, regions, and profits. Analyzing this data helps identify profitable products, loss-making categories, customer behavior, regional performance, and sales trends.

This project focuses on analyzing retail transaction data to understand overall business performance and profitability. The analysis combines **SQL, Python, and Tableau** to clean, analyze, visualize, and present meaningful business insights. The project also focuses on identifying categories and products that contribute to lower profitability and understanding seasonal and regional sales patterns.

2. Abstract

The **Retail Business Performance & Profitability Analysis** project analyzes retail sales transaction data to identify important patterns in sales and profit. The dataset was cleaned and analyzed using SQL and Python, followed by interactive visualization in Tableau.

SQL was used for data checking, cleaning, duplicate identification, and business-level aggregation. Python using Pandas was used for exploratory analysis of sales, profit, categories, sub-categories, regions, customers, discounts, and time-based trends. Tableau was used to create an interactive dashboard containing key performance indicators and charts.

The final dashboard provides a consolidated view of **Total Sales, Total Profit, Profit Margin, Orders, Customers, Category Performance, Sub-Category Profit, Regional Performance, Monthly Sales Trends, Top Products, and Discount vs Profit**.

Interactive filters allow the user to explore the results by **Region, Category, Segment, Year, and Season**.

3. Tools Used

- **SQL / SQLite** – Data checking, cleaning, duplicate detection, and aggregation
- **Python** – Data analysis and exploratory data analysis
- **Pandas** – Data manipulation and analysis
- **Matplotlib / Seaborn** – Data visualization
- **Tableau** – Interactive dashboard creation
- **Jupyter Notebook** – Python analysis environment

The project guidelines identify SQL, Python/Pandas, Seaborn, and Tableau as the main tools for this project.

4. Steps Involved in Building the Project

Step 1: Data Collection and Preparation

The retail transaction dataset was imported and examined to understand its structure and available fields.

Step 2: Data Cleaning using SQL

SQL queries were used to check the number of records, identify missing values and duplicate records, and create a cleaned orders_clean table.

Step 3: SQL Business Analysis

SQL queries were created to calculate overall sales, profit, profit margin, quantity, orders, customers, category performance, sub-category performance, regional performance, segment performance, discount analysis, yearly and monthly performance, product performance, and customer performance.

Step 4: Exploratory Data Analysis using Python

Python and Pandas were used to analyze the cleaned data and identify patterns in sales, profit, categories, sub-categories, regions, customers, discounts, returns, and time-based performance.

Step 5: Tableau Visualization

The cleaned retail data was connected to Tableau. KPI cards and charts were created for sales, profit, profit margin, orders, customers, categories, sub-categories, regions, monthly trends, products, and discounts.

Step 6: Interactive Dashboard

An interactive dashboard was created with filters for **Region, Category, Segment, Year, and Season**, allowing users to explore different parts of the retail data.

5. Conclusion

The project demonstrates how retail transaction data can be transformed into useful business insights using SQL, Python, and Tableau. The analysis provides a consolidated view of sales and profitability across products, categories, regions, customers, and time periods.

The Tableau dashboard makes the results easier to understand through interactive visualizations and filters. The project can help businesses identify areas requiring attention, understand profitability patterns, and support data-driven analysis of retail performance.